

Stereo Depth Estimation Using OpenCV

Based on the OpenCV Aloe Stereo Pair and StereoSGBM

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01 Experiment Overview

This experiment implements binocular disparity estimation and relative-depth visualization with OpenCV. The official OpenCV aloe stereo pair is used as the input. StereoSGBM matches corresponding pixels, and the fixed-point output is converted to pixel disparity for visualization.

The execution completed successfully on images of 1282 x 1110 pixels. The positive-disparity ratio was 78.07%. It expresses the coverage of usable disparity rather than an accuracy measurement.

Stereo Pair	Image Size	Valid Disparity	Conclusion
AloeL / AloeR	1282 x 1110 px	78.07%	Successful

02 Principle of Stereo Depth Estimation

After stereo rectification, corresponding points lie on the same horizontal line. Their horizontal displacement is the disparity d . With focal length f and baseline B , metric depth is $Z = fB / d$. Nearby objects therefore have larger disparity, while distant objects have smaller disparity.

The supplied pair is already suitable for matching, but calibrated focal length and baseline are not used here. The program therefore displays relative depth $1/d$ rather than distance in metres. Zero or negative disparity is invalid and is displayed in black.

03 Experimental Environment and Parameters

Item	Configuration
Programming Language	Python 3.8
Vision Library	OpenCV 4.12.0
Numerical Library	NumPy 1.24.4
Input Images	OpenCV aloeL.jpg and aloeR.jpg
Stereo Method	StereoSGBM 3-way
Number of Disparities	128
Block Size	5 x 5

04 Experimental Method

4.1 Input Preparation

The script downloads `aloeL.jpg` and `aloeR.jpg` from the official OpenCV sample-data repository into `data/`. The images are two views of the same static scene and have identical dimensions.



Figure 1: Left image from the OpenCV aloe stereo pair

4.2 Disparity Computation

Both images are converted to grayscale. StereoSGBM uses 128 disparities, a 5 x 5 matching window, uniqueness filtering, and speckle filtering. The returned fixed-point disparity is divided by 16.0 to obtain pixel disparity.

4.3 Relative-depth Visualization

For valid pixels, relative depth is computed as $1 / \text{disparity}$. The 2nd and 98th percentiles normalize the displayed range. A TURBO colour map makes changes in disparity and relative depth easier to inspect.

Execution command: `python stereo_depth.py`

05 Experimental Results

Metric	Result
Image Size	1282 x 1110 px
Number of Disparities	128
Block Size	5
Valid Disparity Ratio	78.07%
Generated Output Files	7

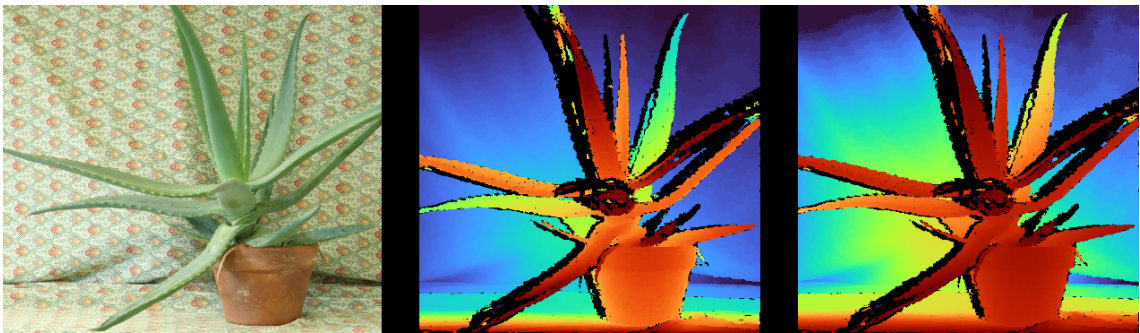


Figure 2: Left image, disparity visualization, and relative-depth visualization

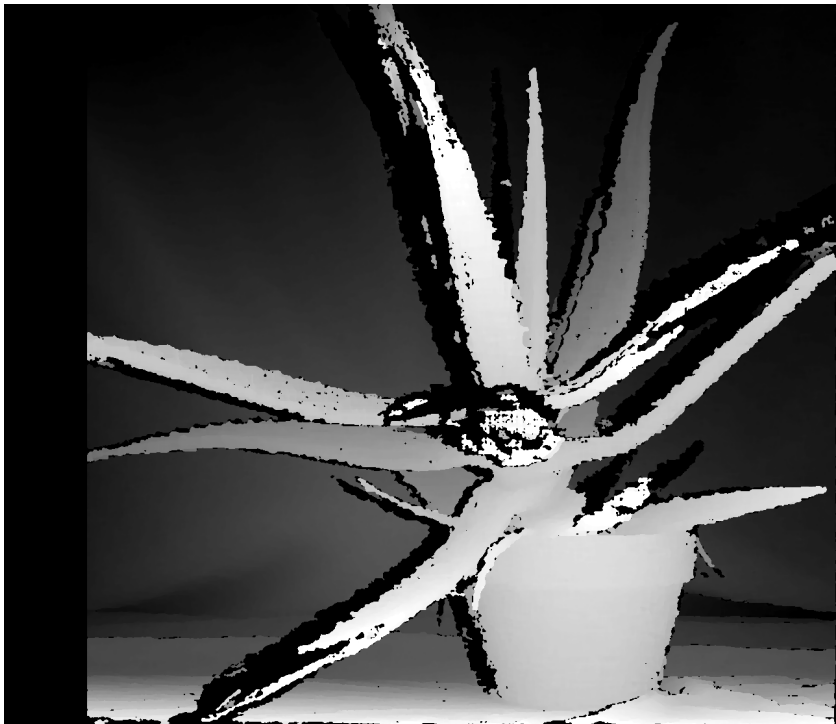


Figure 3: Normalized grayscale disparity map

06 Analysis and Discussion

The plant leaves and flowerpot are estimated as closer than the patterned background because their disparities are larger. These regions have clear boundaries and rich texture, which supports reliable correspondence matching. The background has smaller disparity and appears in a different colour range.

Black regions occur where matching is unreliable, especially at occlusion boundaries, narrow leaf edges, and weakly textured areas. A pixel visible in one camera may be hidden in the other, so a valid correspondence is not always available. Repeated patterns can also create ambiguous matches.

A calibrated stereo camera would enable metric depth using focal length and baseline. Further improvement would include camera calibration, stereo rectification, and parameter tuning for the particular scene. This result is appropriate for comparing relative distance in the image pair, but not for reporting physical distance.

07 Conclusion

This experiment successfully implemented OpenCV-based stereo depth estimation with the official aloe stereo data. StereoSGBM produced a valid disparity ratio of 78.07%, and the program exported disparity images, a relative-depth image, raw disparity data, and a comparison panel. The result demonstrates the inverse relationship between disparity and scene depth.

08 Submission Record

Item	Record
GitHub Repository	https://github.com/Eva122205/opencv-stereo-depth
Program Commit	bb82872 - Add StereoSGBM stereo depth estimation program
Results Commit	739245f - Add stereo disparity and relative depth results
Report Commit	Added after report generation

The program, experimentally generated results, and report are committed in separate stages so the repository history records the work clearly.

09 References

- [1] OpenCV Documentation. StereoSGBM Class Reference.
- [2] OpenCV Sample Data Repository. aloeL.jpg and aloeR.jpg stereo pair.
- [3] H. Hirschmuller. Stereo Processing by Semiglobal Matching and Mutual Information. IEEE TPAMI, 2008.